Name:	Class:	Date given:
		Date due in:

Photosynthesis

1. What is the purpose of photosynthesis? [2]

_____

2a. Complete the word equation for photosynthesis. [2]

carbon dioxide + _____ $\xrightarrow[Y]{X}$ _____ + oxygen

2b. What are the parts labelled X and Y on the word equation? [2]

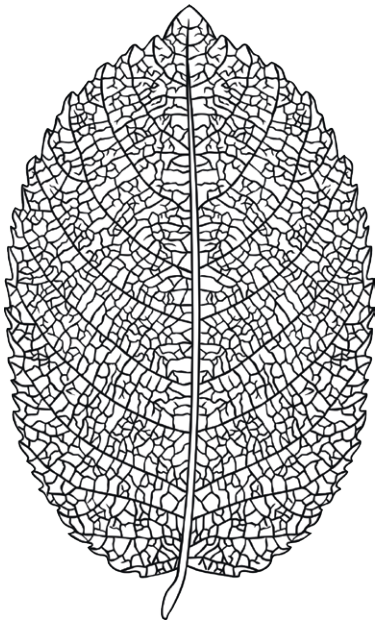
_____

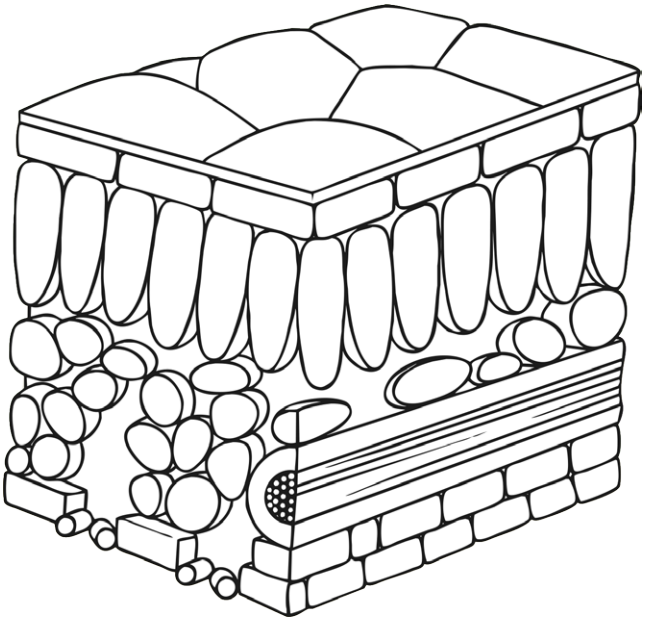
3. Match the part of the plant with its function, by drawing a line to connect the boxes. [2]

leaves	transports glucose and water within the plant
roots	contains chlorophyll for photosynthesis
stem	absorbs minerals and water from the soil

twinkl.co.uk

Look at the diagrams below. There is a typical plant leaf and a magnified cross section of part of the leaf.





4. Describe how a leaf is adapted to carry out photosynthesis efficiently. [4]

Learning Outcomes (tick if achieved)

Q1	Describe photosynthesis as a process for plant nutrition.	
Q2	Describe photosynthesis in the form of a word equation.	
Q3	Describe structure and function of the parts of a plant.	
Q4	Explain adaptations of leaves for photosynthesis.	

twinkl.co.uk

Photosynthesis **Answers**

1. What is the purpose of photosynthesis? [2]

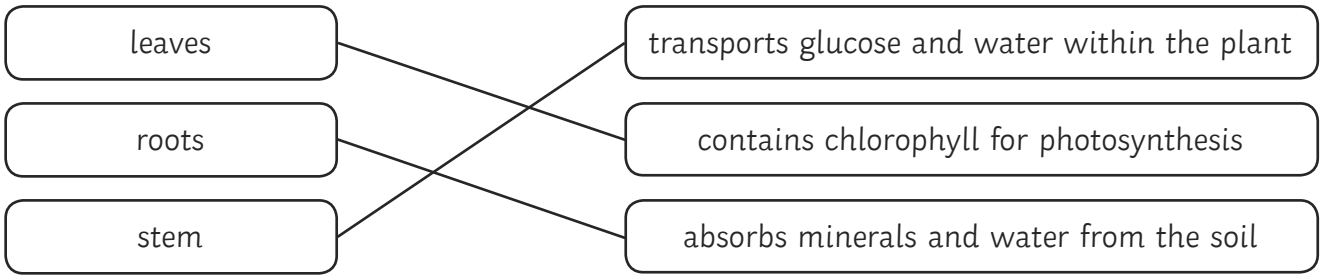
1 from - produce food / glucose (reject energy)
- for growth / increase biomass

carbon dioxide + **water** $\xrightarrow[Y]{X}$ **glucose** + oxygen

2b. What are the parts labelled X and Y on the word equation? [2]

sunlight, chlorophyll (any order)

3. Match the part of the plant with its function, by drawing a line to connect the boxes. [2]



4. Describe how a leaf is adapted to carry out photosynthesis efficiently. [4]

Max 4 marks; 1 mark for each:

large surface area / broad shape
Absorb more sunlight
Chloroplasts
Chloroplasts at top of leaf
(therefore) more sunlight
Stomata / tiny holes
Stomata on underside of leaf
Stomata for gas exchange
Air spaces inside leaves
Gases move easily between cells
Contain network of veins
Xylem
Phloem
Transports water (to) and sugar (from) leaves